# Matched $V_2$ and $d4$-readout decay closes the scale escape but erases the compositional circuit

8 $\alpha$ × 8 $\lambda$ × 3 seeds = 192 cells

## Design

This control is identical to the V2-only-decay experiment except that the same phase-2 coefficient $\lambda$ penalizes both $V_2$ and the $d4$ readout row $W[2]$. The embedding, $V_1$, and the two $d2$ readout rows remain decay-free; all parameters remain trainable and the two $d2$ losses remain active. The initial accessibility is set by $V_1 \mapsto \alpha V_1$ and measured as $R_0 = Q_2/Q_1$. The grid uses eight $\alpha$ values, eight $\lambda$ values from 0 to $10^{-1}$, and three pretrained seeds. All 192 cells completed, retained $d4$ accuracy, and are represented in the trajectory files.

## Findings

**1. Readout decay does not sharpen the transition at $R_c = 1/\sqrt{8}$.** Very weak decay gives a small increase in composition, but the effect reverses before a phase develops. At $R_0 = 0.417$, the mean 2+2 fraction is 0.101 ($\lambda = 0$) → 0.124 ($3{\cdot}10^{-4}$), then falls to 0.015 ($10^{-3}$) and 0.0028 ($3{\cdot}10^{-3}$). At $R_0 = 0.515$ the corresponding sequence is 0.103 → 0.128 → 0.024 → 0.0021. Pooling all supercritical cells gives $f_{2+2} = 0.112$ at $3{\cdot}10^{-4}$, 0.020 at $10^{-3}$, and 0.0021 at $3{\cdot}10^{-3}$. The phase diagrams therefore become uniformly blue as $\lambda$ grows; there is no decay-sharpened boundary.

**2. The contrast with both benchmarks is large.** At $R_0 = 0.417, \lambda = 3{\cdot}10^{-3}$, matched readout decay gives $f_{2+2} = 0.0028$, compared with 0.185 under V2-only decay. At $\lambda = 10^{-2}$ the comparison is 0.0007 versus 0.159. The frozen benchmark is more strongly compositional still: around matched $R \simeq 0.436$, it gives $f_{2+2} = 0.84$–$0.89$ at $3{\cdot}10^{-3}$ and 0.87–0.92 at $10^{-2}$. Readout decay thus does not complete the V2-only experiment's partial transition; it removes it.

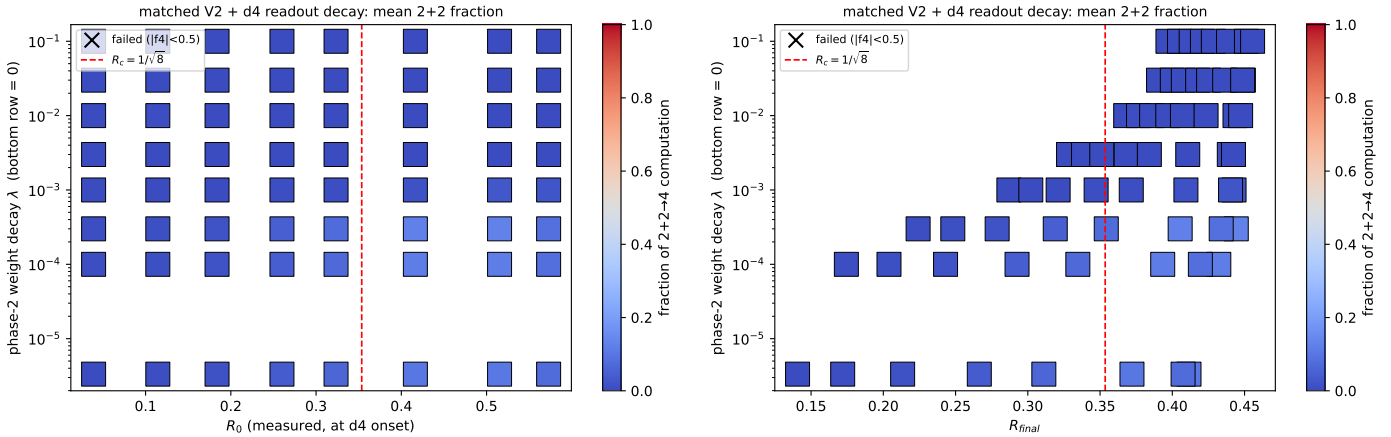



Figure 1: Mean compositional fraction versus $R_0$ (left) and $R_{\text{final}}$ (right). The red line marks $R_c$. Weak decay preserves a faint accessibility dependence, but $\lambda \geq 10^{-3}$ suppresses composition on both sides of the theoretical boundary.

**3. $R(t)$ remains high while composition disappears.** Decay raises $R_{\text{final}}$ relative to the no-decay trajectory and compresses the sweep toward a supercritical band. For example, $R_0 = 0.323$ ends at $R_{\text{final}} = 0.311$ with no decay but at 0.407 for $\lambda = 10^{-2}$; $R_0 = 0.515$ ends at 0.412 and 0.443, respectively. High-$R_0$ conditions can drift downward in absolute terms, while low-$R_0$ conditions drift upward. The important observation is the decoupling: at $\lambda = 10^{-2}$ the representation is mostly above $R_c$, yet the compositional fraction is only $4{\cdot}10^{-4}$ in the initially supercritical group.

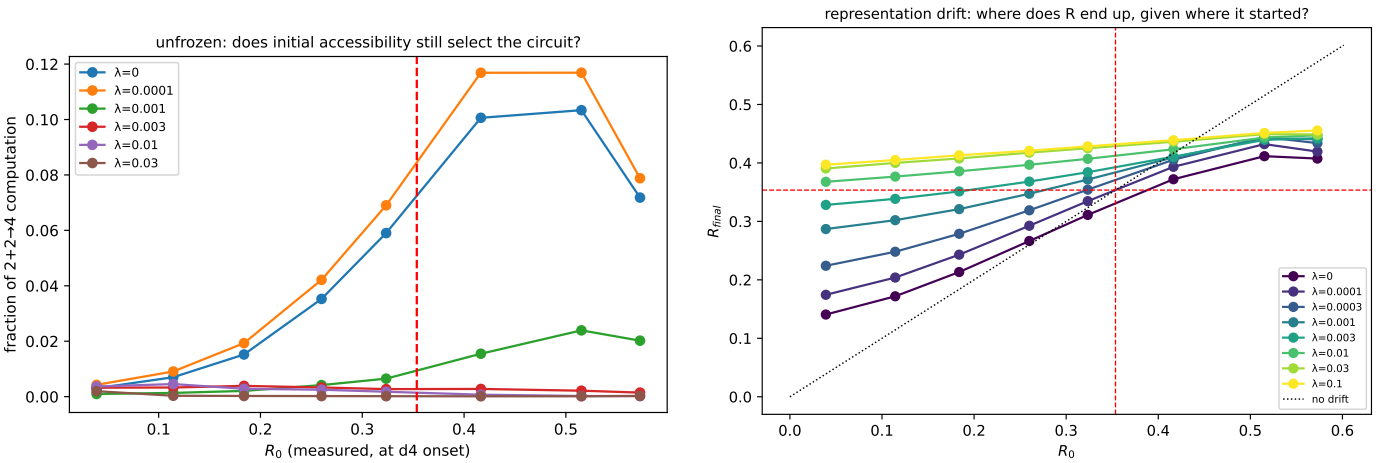



Figure 2: Left: fixed-$\lambda$ cuts show the small weak-decay separation and its collapse at larger $\lambda$. Right: $R_{\text{final}}$ versus $R_0$; decay funnels the representation toward $R \approx 0.4$–$0.45$ rather than restoring a compositional phase.

**4. The V2-only scaling escape is fixed, but balanced norms do not emerge.** In the V2-only experiment, $V_2$ could vanish while the unpenalized readout stayed large: at $R_0 = 0.417, \lambda = 10^{-2}$, $\|V_2\| = 4.7{\cdot}10^{-4}$ while the total readout norm was 2.83. With matched decay, both relevant norms shrink: at the same $R_0$, $(\|V_2\|, \|W_4\|) = (0.420, 0.734)$ at $10^{-3}$, $(0.116, 0.372)$ at $10^{-2}$, and $(0.067, 0.276)$ at $3{\cdot}10^{-2}$. However, the ratio $\|W_4\|/\|V_2\|$ grows from 1.75 to 3.22 to 4.11 rather than approaching the near-balanced frozen solution (roughly 1.8 versus 1.9 around the same $R$). Matched parameter decay therefore removes the literal rescaling loophole but does not select the frozen compositional optimum in the full network.

**5. The first-block residual shortcut still blocks a pure compositional phase.** For $\lambda \leq 3{\cdot}10^{-2}$, $|\hat{f}_4|$ remains at least 0.998 even where $V_2$ is tiny and $f_{2+2}$ is nearly zero. Hundreds of block-2 neurons remain classified as relevant, but their computation is not parent-compositional. Because the unpenalized, trainable first block feeds the readout through the residual stream, the most consistent interpretation is that its learned $d4$ leak carries the target while decay dismantles the two-parent block-2 route. This is an inference from the present observables rather than a direct route decomposition; the parallel-$d4$-adapter experiment records first- and second-block contributions explicitly.

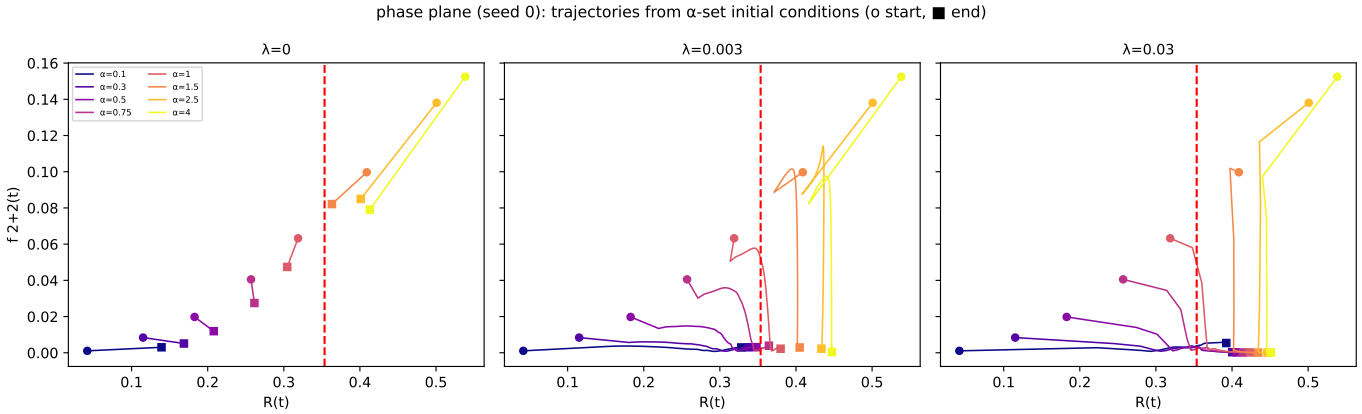



Figure 3: Seed-0 trajectories in $(R(t), f_{2+2}(t))$. Under appreciable matched decay, trajectories move toward or remain in the supercritical-$R$ region while falling toward zero composition. Accessibility alone no longer predicts which circuit carries $d4$.

**Bottom line.** Readout decay answers the scale-balance objection to V2-only decay, but the result is not a sharper transition. Instead, it reveals a stronger bypass: the full network keeps $d4$ solved, maintains or increases measured accessibility, and abandons the 2+2 circuit. The frozen benchmark's clean transition requires not only matched regularization but also protection from alternative trainable routes. Directly regularizing a new block-1 $d4$ adapter alongside $V_2$ and $W_4$, while freezing the pretrained $d2$ parents, is therefore the appropriate next comparison.

Artifacts: `analysis_output/accessibility_alpha_v2w4wd/` (merged results, trajectories, phase diagrams, cuts, drift, and phase-plane plot). Code: `analysis/accessibility/run_alpha_unfrozen_experiment.py` (`--wd-block2-readout`) and `scripts/launch_v2w4wd_workers.sh`.